

Arthroscopic Dynamic Anterior Stabilization and Augmentation of Bony Bankart Lesions With Transfer of the Long Head of the Biceps

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Abstract: Bony Bankart fractures treated arthroscopically with conventional suture anchor fixation are often complicated by fragment displacement and nonunion because of inadequate reduction. This technical report introduces a novel technique using the long head of the biceps (LHB) tendon transfer for elevating bony fragment and augmenting suture anchor fixation. By repositioning the LHB beneath the glenoid fragment, the tendon acts as a dynamic sling, stabilizing the bony fragment during suture anchor fixation. The procedure involves diagnostic arthroscopy, LHB tenotomy and braiding, fragment preparation, glenoid tunnel creation, introducing a guiding suture, LHB transfer for elevating the fragment, and suture anchor fixation for labrum repair. This technique merges assisted reduction with sling stabilization, offering a promising solution for bony Bankart fracture. Further clinical validation is required to confirm the long-term efficacy.

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Bony Bankart fractures, typically caused by traumatic shoulder dislocation or direct impact, particularly those involving the anterior-inferior quadrant, pose significant challenges in arthroscopic management.¹⁻³ Many authors reported secure reduction and fixation of the Bankart fracture along with capsular repair.⁴⁻⁶ However, traditional suture anchor fixation techniques often struggle with maintaining reduction because of poor visualization, medial sagging of the fragment, and inadequate stabilization. These limitations increase risks of nonunion, malunion, and post-traumatic osteoarthritis.⁷⁻⁹ The long head of the biceps (LHB) tendon, with its proximity to the glenoid and dynamic biomechanical properties,¹⁰⁻¹² may offer a solution. By transferring the LHB beneath the fragment from the superior side

of the glenoid, surgeons can support reduction and enhance stabilization during arthroscopic labrum repair. The LHB transfer indications for bony Bankart fracture are competitive athletes under 45 and patients with epilepsy, as well as patients aged 45 or above with concomitant superior labrum anterior to posterior lesions (type II or IV; Table 1).

SURGICAL TECHNIQUE

Patient Position, Portal Placement, and Arthroscopic Examination

The patient was placed in the lateral position with the right arm under traction using a 10-pound weight at a 45°

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VIDEO 1 The surgical procedures of arthroscopic dynamic anterior stabilization and augmentation of bony Bankart lesions with transfer of the long head of the biceps. Video content can be viewed at <https://doi.org/10.1002/atn2.70251>.

TABLE 1 LHB Transfer Indications for Bony Bankart Fracture

Indications	1. Younger than 45 years of age participating in competitive sports
	2. Patients with epilepsy
	3. Patients aged 45 or above with concomitant SLAP lesions (type II or IV)

LHB, long head of the biceps.

abduction angle (Video 1). Standard posterior portal was created to perform shoulder examination to detect glenoid fragment; then, anterosuperior portal, anterolateral, anterodistal, and anterior portal (Figure 1) were established (Table 2).

LHB Tenotomy and Braiding

Under observation from the posterior portal, the proximal LHB was ligated by a folded no. 2 polyethylene suture

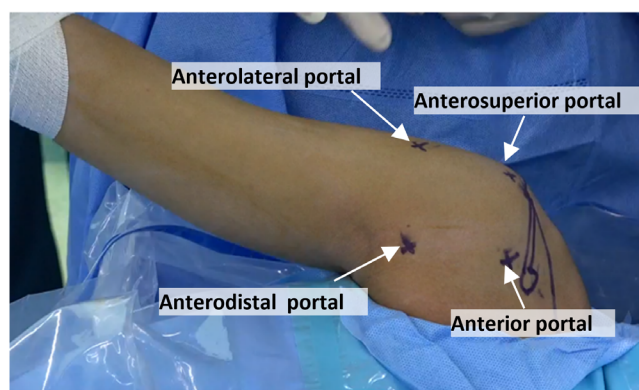


FIGURE 1 View of lateral position with the right arm to show the anterior, anterosuperior, anterolateral, and anterodistal portals.

TABLE 2 Surgical Procedures

1. The patient was positioned laterally with the arm under 10-pound traction at 45° abduction for the procedure.
2. Multiple arthroscopic portals were established to examine the shoulder and locate the glenoid fragment.
3. The proximal long head of the biceps was ligated and released from its superior labral attachment.
4. The LHB was retrieved, braided with an adjustable loop button, and its final diameter was measured.
5. The glenoid fracture was assessed, the hematoma debrided, and the capsular-labral complex was reduced.
6. A customized guide facilitated the creation of a bone tunnel beneath the glenoid fragment.
7. This tunnel was sequentially drilled to match the diameter of the prepared LHB.
8. A guiding suture was passed through the tunnel and the subscapularis tendon for LHB passage.
9. The braided LHB was pulled into the glenoid tunnel using the guiding suture for fracture elevation.
10. The labral fragment complex was repaired with suture anchors, and final tension was applied to the adjustable loop.
LHB, long head of the biceps.

through the anterosuperior working portal, and then, the LHB was detached from the superior labrum using an arthroscopic radiofrequency ablator. The arthroscope is placed in the anterosuperior portal, and then, the anterolateral portal is then created. An obturator was used to expose the distal biceps tendon located at the superior border of the pectoralis major via the anterolateral portal. The proximal end of biceps tendon was retrieved out of anterodistal portal through a string. The biceps tendon and an adjustable loop plate (RIGIDLOOP Adjustable Cortical System; DePuy Synthes, Neuchatel, Switzerland) were braided together with high-strength sutures (Ultrabraid; Smith-Nephew, Andover, MA) (Figure 2). Then, the diameter of the braided proximal LHB was measured.

Fragment Debridement and Preparation

Under observation from the anterosuperior portal, the morphology of Bankart fracture was assessed, medial displacement of the glenoid fragment and collapse of the articular surface margin were found, and then, hematoma was debrided within the shoulder joint cavity from the anterior portal. The capsular-labrum-fragment complex was liberated and reduced to achieve the superior position relative to the glenoid articular plane (Figure 3).

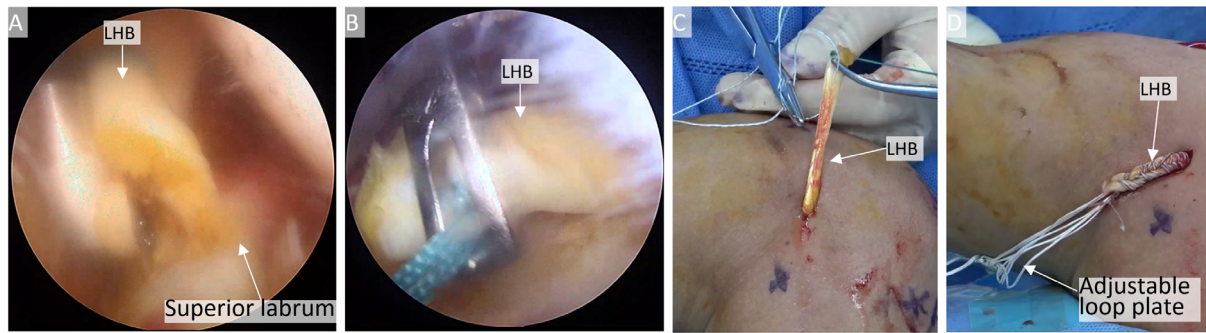


FIGURE 2 The process of LHB preparation of the right shoulder. (A) Detaching the LHB from the superior labrum; (B) extracting the LHB from the anterodistal portal; and (C,D) braiding the LHB with an adjustable loop plate. (LHB, long head of the biceps.)

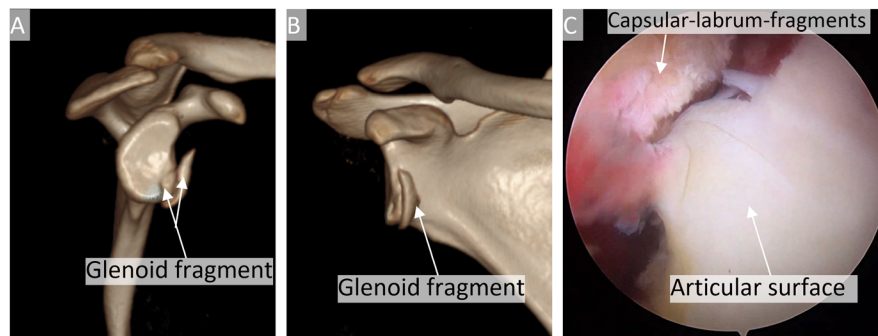


FIGURE 3 The 3-dimensional CT (A,B) and arthroscopic (C) view of glenoid rim fracture of the right shoulder from the anterosuperior portal. (CT, computed tomography.)

Glenoid Tunnel Creation

Under anterosuperior portal arthroscope observation, a customized glenoid guide was sticking to the articular surface via the posterior portal. A 2.5-mm Kirschner wire was drilled into the posterior glenoid surface positioned 8-9 o'clock and advanced to the anterior exit point. The location of the bone tunnel exit is determined by the position of the fracture fragment, positioned at the midpoint of the fracture line in the sagittal plane and proximal to the fracture edge in the transverse plane. This Kirschner wire pathway was subsequently expanded using a 4.5-mm cannulated drill to establish the primary glenoid tunnel. A 5.0-mm retrograde drill matching the diameter of proximal LHB was used to enlarge the anterior tunnel segment (Figure 4). The intraosseous portion of LHB generally spanned 5 mm.

Introducing Guiding Suture

Under arthroscopic visualization via the anterosuperior portal, a folded high-strength suture (the guiding suture) was introduced through the glenoid tunnel in a posterior-to-anterior tunnel using a suture passer. The guide suture

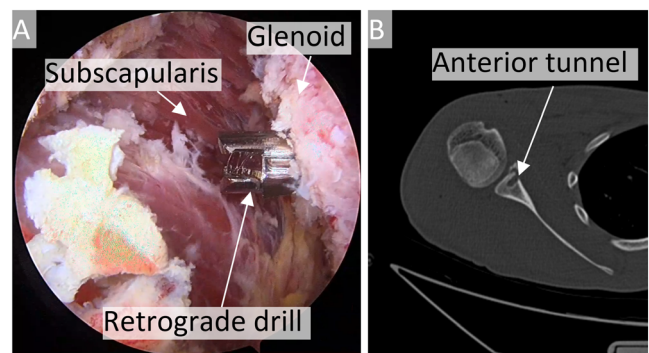


FIGURE 4 Illustration of arthroscopic (A) and CT transverse view (B) of the right shoulder to show the location of the glenoid tunnel from the anterosuperior portal. (CT, computed tomography.)

was subsequently retrieved into the joint cavity using a grasping instrument. The initial suture passer was redirected through the posterior portal to re-engage the guiding suture. This suture was then advanced anteriorly beneath the fracture fragment, penetrating the subscapularis tendon at the 4:30 clock position. The arthroscope was repositioned through the anterolateral portal and advanced along the subscapularis' anterior aspect. Another suture passer introduced through the anterosuperior portal facilitated fibrous tissue

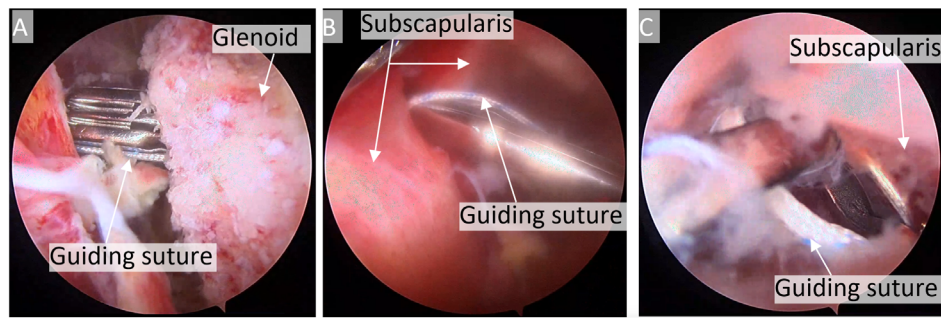


FIGURE 5 The pathway of guiding suture at the glenoid tunnel (A), posterior (B), and anterior (C) side of the subscapularis muscle of the right shoulder from the anterosuperior and anterolateral arthroscopic views.

dissection until making contact with the subscapularis-penetrating suture passer (Figure 5). This guiding suture was then exteriorized through the anterosuperior portal.

LHB Transfer for Elevating Fragment

The prepared LHB braided with adjustable loop plate was threaded through a cannula advanced into the anterior subdeltoid space via the anterodistal portal. A specialized grasper within this cannula retrieved the guiding suture. Subsequent traction on the guiding suture facilitated passage of the adjustable loop plate sutures through both the subscapularis and glenoid tunnels, achieving posterior portal exteriorization. Final arthroscopic confirmation through the anterosuperior portal verified that the LHB was pulled to the anterior portion of the glenoid tunnel (Figure 6) for elevating the fragment via tensioning adjustable loop.

Suture Anchor Fixation for Labrum Repair

Under posterior portal for observation, two anchor placement sites were created via anterior portal at the 3

and 6 o'clock positions along the anterior border of the glenoid fracture. A single guiding suture was threaded through the capsule-labral complex at the 6 o'clock position. Another guiding suture was sequentially placed through the superior capsule-labral complex tissue at the 3 o'clock positions. The 6 o'clock anchor (SutureTak Anchor; Arthrex, Naples, FL) was implanted first, with its associated sutures exteriorized through the posterior portal. The 3 o'clock anchor (SutureTak Anchor; Arthrex, Naples, FL) follows, with its sutures routed through the anterosuperior portal. Before using the pre-positioned labral guiding suture, the capsule-labral fragment complex was firstly repositioned above the LHB to obtain reduction and, then, attached to the anterior edge of the glenoid bed by sutures from the 3 and 6 o'clock anchors (Figure 7). Finally, the adjustable loop plate was further drawn to the glenoid (Figure 8).

DISCUSSION

Bony Bankart fractures, often associated with traumatic anterior shoulder instability, pose significant challenges in orthopedic surgery. These fractures, classified as Ideberg type Ia fractures, account for 4%-70% of cases after shoulder dislocations.³ Over the past decade, arthroscopic techniques have evolved dramatically, offering minimally invasive solutions with improved precision in fragment reduction, fixation, and soft tissue repair.^{5,13,14} Bony Bankart fractures, particularly anterior-inferior fractures, are challenging to manage arthroscopically because of fragment instability and suboptimal reduction.

Traditional suture anchor fixation techniques often struggle with medial sagging of the fracture fragment, leading to nonanatomic reduction, residual articular step-off, and increased risks of nonunion or post-traumatic arthritis.^{8,15} Fragment displacement persists in 15%-30% of cases because of gravity and inadequate soft tissue support, leading to nonunion rates of 10%-20%.⁹ Even the technique of

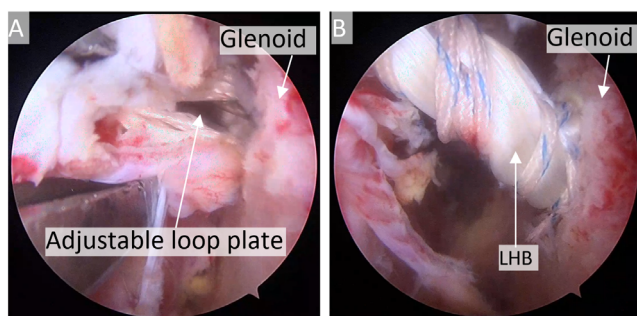


FIGURE 6 The passage of adjustable loop (A) and LHB (B) at the glenoid tunnel of the right shoulder from the anterosuperior arthroscopic view. (LHB, long head of the biceps.)

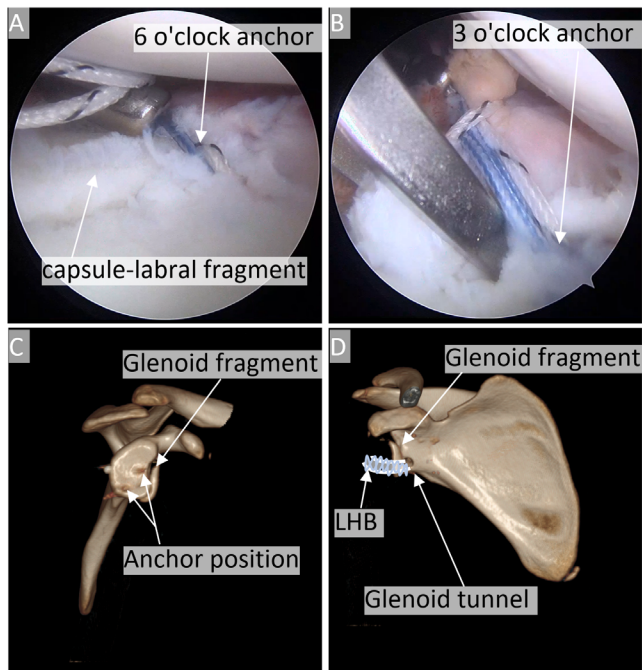


FIGURE 7 Reattaching the capsule-labral complex to the glenoid edge of the right shoulder with 6 (A) and 3 (B) o'clock anchors from posterior arthroscopic view. The sagittal (C) and coronal (D) view of CT to show the reduction of the capsule-labral fragment. (CT, computed tomography; LHB, long head of the biceps.)

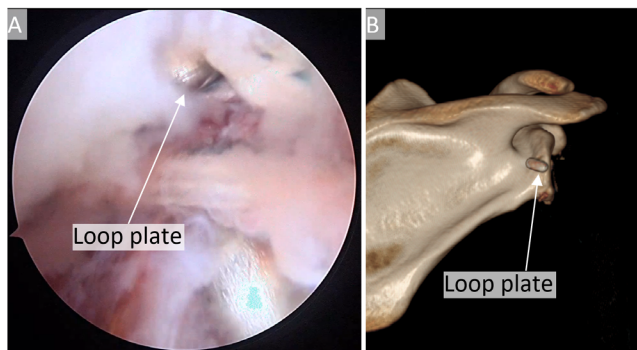


FIGURE 8 The location of loop plate from arthroscopic (A) and CT (B) view of the right shoulder. (CT, computed tomography.)

double-row bridge fixation involves placing medial anchors in the glenoid neck and lateral anchors at the fragment edge, compressing the fragment anatomically. The study still exhibits 10% nonunion rates.¹⁶ The LHB transfer technique addresses these shortcomings by providing mechanical advantages,^{10,17} especially when dealing with large bony fragments. Positioning the LHB beneath the fragment creates a “hammock effect,” during the process of fragment reduction and fixation, counteracting posterior sagging and enabling anatomic alignment. Furthermore, during shoulder motion, the tensioned LHB dynamically stabilizes the fragment, mimicking the native glenoid curvature,¹² resisting medial displacement. This biomechanical support is critical for maintaining reduction during early rehabilitation. The LHB’s vascularity may enhance fracture healing, reducing nonunion rates. The 6-month postoperative computed tomography scan findings for one of our patients align with the aforementioned theoretical advantages, showing remodeled healing of the Bankart fragment, no medial displacement of the bone fragment, and restoration of the glenoid curvature (Figure 9).

Restoration of the glenoid arc is paramount for preventing instability and osteoarthritis.^{18,19} By preserving the curvature, this technique optimizes dynamic bone remodeling biomechanics, offering a functional advantage over traditional methods. However, this technique may have the following limitations: The first is requiring advanced arthroscopic skills. The second is that sacrifice of the LHB may theoretically affect biceps function, though clinical impact remains debated. Finally, LHB transfer has a risk of compromising the axillary nerve during the process of subscapularis penetration. The advantages and disadvantages of this procedure are summarized in Table 3.

Arthroscopic LHB transfer with suture anchor fixation may be a promising advancement in managing displaced bony Bankart fracture. It provides adequate reduction, and dynamic stabilization may be a favorable option in

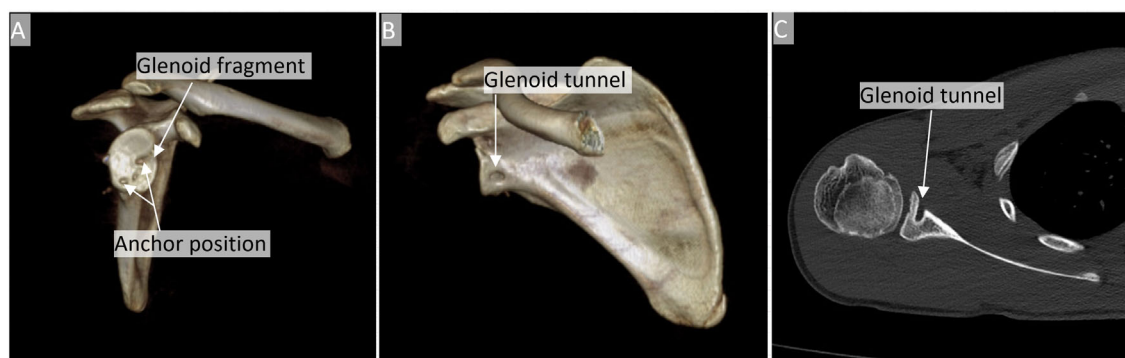


FIGURE 9 The 3-dimensional sagittal (A), coronal (B), and axial view of 6-month CT to show the healing of glenoid fragment of the right shoulder. (CT, computed tomography.)

TABLE 3 Advantages and Disadvantages

Advantages
1. Improved reduction: LHB directly elevated the fragment and blocking medial movement of capsule-labral fragment during the process of capsule-labrum repairing.
2. Augmenting construct: Integration of the LHB into the suture anchor construct reinforced resistance to shear forces.
3. Dynamic stabilization: The LHB acts as a sling, resisting medial sagging during shoulder motion.
4. Reduced nonunion risk: Enhanced fragment compression with a tension-band from the LHB.
Disadvantages
1. Technically demanding: Requiring advanced arthroscopic skills.
2. Sacrifice of the LHB: Theoretically affecting biceps function, though clinical impact remains debated.
LHB, long head of the biceps.

TABLE 4 Pearls and Pitfalls

1. Fragment preparation: The capsule-labral fragment complex was liberated enough to achieve superior fracture reduction.
2. Tunnel engineering specifications: Calculate osseous tunnel length based on glenoid width measurements, preserving a 5-mm posterior cortical bridge. Avoid full-tunnel expansion to maintain miniplate suspension feasibility.
3. Safe subscapularis penetration zone: The penetrating point of subscapularis was the 4:30 position, avoiding axillary nerve compromise.
4. Fixation protocol: When adjustable loop plate was tensioned, ensuring LHB translocation into the osseous tunnel.
LHB, long head of the biceps.

competitive athletes under 45 and patients with epilepsy, as well as patients aged 45 or above with concomitant superior labrum anterior to posterior lesions (type II or IV). The pearls and pitfalls of this procedure are illustrated in Table 4. Further follow-up clinical studies are warranted to validate long-term outcomes and refine the technical nuances.

DISCLOSURES

The author (G.X.) declares the following financial interests/personal relationships which may be considered as potential competing interests: G.X. report a relationship with the Natural Science Foundation of Shanghai Municipality that includes funding grants. The other authors (W.S., X.W., J.Z.) declare that they have no known competing

financial interests or personal relationship that could have appeared to influence the work reported in this article.

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